

Gretchen Lam

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EDUCATION

University of California, Santa Barbara

Bachelor of Science in Computer Science | GPA: 3.73 / 4.00

Relevant Coursework: Object Oriented Design, Data Structures and Algorithms, Computer Architecture, Artificial Intelligence, Machine Learning, Computer Vision, Advanced Application Programming, Human-Computer Interaction

Santa Barbara, CA

Sep. 2021 - March 2025

EXPERIENCE

Software Engineer

April 2025 - Present

Lockheed Martin AI Center — Remote

Active Security Clearance: Top Secret

- Designed and implemented frontend capabilities for STAR.UI™, an AI-enabled operational platform, enabling real-time visualization, monitoring, and interaction with complex mission systems.
- Developed and maintained frontend applications using Svelte, React, Next.js, and TypeScript for internal operational and engineering programs, including geospatial visualization and mapping interfaces.
- Contributed to agentic AI tooling workflows for Unreal Engine asset selection and environment interaction, supporting AI-assisted simulation and content generation processes.
- Collaborated with multidisciplinary engineering teams to integrate frontend applications with backend APIs, real-time data services, and containerized software environments.

Software Engineer Intern

June 2024 - Sep. 2024

Raytheon — Goleta, CA

- Contributed to the development and enhancement of the Keysight EW Threat Simulation View application using C# and .NET in the MVVM design pattern, optimizing system responsiveness through multithreading
- Addressed cross-platform compatibility issues to ensure consistent software behavior across various hardware environments and improved the efficiency of data processing features
- Developed firmware to support the hardware capabilities of a microcontroller-based system, including custom communication protocols for device interfacing, enhancing overall system integration
- Improved system robustness by implementing monitoring features and fail-safes, significantly boosting the reliability and fault tolerance of the embedded system

Software Engineer Intern

June 2023 - Sep. 2023

Arthrex — Santa Barbara, CA

- Designed and implemented a dynamic GUI using Python's Tkinter library to automate the provisioning of up to 200 emulated imaging devices simultaneously, streamlining their integration with an IoT platform and site server
- Optimized backend infrastructure by containerizing services with Docker, significantly reducing the overhead of setting up and managing device emulations and improving performance scalability
- Executed over 500 automated test cases on emulated imaging devices using Python's Selenium, validating core functionalities such as image and video capture, upload, and transmission
- Utilized the SpecFlow BDD framework for .NET to write comprehensive, automated test scripts, ensuring thorough validation of business logic and enhancing software quality with faster, continuous integration cycles

PROJECTS

Closet Tracker

Jan. 2025 - Mar. 2025

- Developed a cross-platform React Native mobile application using Expo Go and Firebase, allowing users to digitally organize their wardrobe by uploading, categorizing, and managing clothing items within a user-friendly interface
- Implemented an intuitive image upload feature that enables users to capture photos using their device's camera or select images from their gallery, with real-time previews and validation for optimal user experience
- Integrated Supabase as a third-party cloud storage solution to generate public image URLs, which are then stored in Firebase Firestore for seamless retrieval and display of clothing items within the wardrobe interface

Facial Frenzy

Sep. 2023 - Dec. 2023

- Developed a Python-based interactive facial expression game, which involves user authentication, emotion recognition challenges, and dynamic scoring
- Engineered an accurate real-time facial recognition and authentication system using OpenCV and face-recognition libraries in Python, with a dynamic and innovative sign-up process for new users
- Contributed to the development of an emotion recognition gaming feature, enhancing user engagement by challenging players to match facial expressions with randomly generated emotions
- Integrated the game with a Flask web application and PostgreSQL for efficient and secure user data management and Firebase Storage for reliable cloud-based data storage

TECHNICAL SKILLS

Programming Languages: C++, C#, Python, Java, HTML, CSS, JavaScript, Assembly

Software Experience: .NET, Visual Studio, Docker, Unity, React, React Native, Selenium, Linux, Git, Jira, Confluence, Expo Go